

Econ 101A

Section 5

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Fall 2026

We can now find an optimum. This section is about *using* it. Two questions. First, how does the optimal choice move when a parameter (a price, a wage) changes, even when we cannot solve for it in closed form? The tool is the **Implicit Function Theorem**. Second, what is the Lagrange multiplier really? We will *derive* that it equals $\partial V/\partial b$ with the envelope theorem, and build an intuition for it as a shadow price.

1 Comparative Statics with the Implicit Function Theorem

Recall the IFT: if $F(x; p) \equiv 0$ defines x as a function of p , then, wherever $F_x \neq 0$,

$$\frac{dx}{dp} = -\frac{F_p}{F_x}.$$

The trick. The optimal choice $x^*(p)$ is *defined* by the first-order condition. For $\max_x f(x; p)$, the optimum solves $f_x(x^*(p); p) = 0$. This is exactly an equation $F(x; p) = 0$ with $F = f_x$. Apply the IFT:

$$\frac{dx^*}{dp} = -\frac{f_{xp}}{f_{xx}}.$$

The second-order condition says $f_{xx} < 0$ at a maximum, so the denominator is negative and

$$\text{sign}\left(\frac{dx^*}{dp}\right) = \text{sign}(f_{xp}).$$

In words: the optimal action rises with a parameter exactly when that parameter raises the *marginal* payoff of the action. This one line is the engine behind most comparative-statics results, and it never required solving the FOC.

Two choice variables. With $\max_{x,y} f(x, y; p)$, differentiate both FOCs in p to get a linear system whose coefficient matrix is the Hessian H_f ; solving it (Cramer's rule) gives dx^*/dp and dy^*/dp . Same idea, more bookkeeping.

*These notes are a consolidation of the notes of Dan Acland, Ivan Balbuzanov, Mariana Carrera, Justin Gallagher, Anne Karing, Matt Leister, Sam Leone, Nicholas Li, Pablo Munoz, Yassine Sbair Sassi, Jon Schellenberg, Katalin Springel, Jonas Tungodden, David Wu, Sam Wang, and Matteo Saccarola.

2 The Envelope Theorem, Derived

Let $V(b) = \max_{x,y} f(x,y)$ subject to $g(x,y) = b$, with solution $(x^*(b), y^*(b), \lambda(b))$. We show $V'(b) = \lambda$.

Since $V(b) = f(x^*(b), y^*(b))$, the chain rule gives

$$V'(b) = f_x x^{*'} + f_y y^{*'}.$$

At the optimum the FOCs say $f_x = \lambda g_x$ and $f_y = \lambda g_y$, so

$$V'(b) = \lambda(g_x x^{*'} + g_y y^{*'}).$$

Now differentiate the constraint $g(x^*(b), y^*(b)) = b$ with respect to b :

$$g_x x^{*'} + g_y y^{*'} = 1.$$

Substituting,

$$\boxed{V'(b) = \lambda}.$$

The indirect effects through x^* and y^* collapsed because at the optimum the objective and the constraint move together. Shortcut: $V'(b) = \partial \mathcal{L} / \partial b|_{\text{opt}} = \lambda$, since $\mathcal{L} = f + \lambda(b - g)$.

3 What a Multiplier Is

- **A shadow price.** λ is the extra objective you get per extra unit of the constrained resource: the internal price the optimizer would pay for one more unit. In the consumer's problem $b = m$, so $\lambda = \partial V / \partial m$ is the **marginal utility of income**.
- **A common marginal value.** At the optimum, the last unit of the resource yields the *same* marginal payoff, λ , wherever you spend it. If one use paid more, you would move a unit there; the fact that you stopped means the margins are equal. λ is that common margin.
- **Everywhere.** A budget gives the marginal utility of income; a firm's output target gives marginal cost; an hour budget gives the value of an hour; a capacity limit gives the rent you would pay for one more unit of capacity.

4 Exercises

Exercise 1 (IFT: signing a firm's input demand)

A competitive firm chooses an input $x \geq 0$ to maximize profit $\pi(x) = pF(x) - wx$, where $F'(x) > 0$ and $F''(x) < 0$, and $p, w > 0$.

- Write the first-order condition, and explain why for a general F you cannot solve it in closed form.
- Use the IFT to sign $\frac{dx^*}{dw}$ and $\frac{dx^*}{dp}$.
- Interpret, and say where the second-order condition entered.

Exercise 2 (derive $\lambda = \partial V / \partial m$)

A consumer maximizes $u(x, y) = \alpha \ln x + \beta \ln y$ subject to $p_x x + p_y y = m$ (with $\alpha, \beta > 0$).

- (a) Recall the demands and the multiplier from Section 3.
- (b) Write the indirect utility $V(p_x, p_y, m) = u(x^*, y^*)$ and show it has the form $(\alpha + \beta) \ln m + C(p_x, p_y)$, where C does not depend on m .
- (c) Compute $\partial V / \partial m$ and check it equals λ .
- (d) Interpret λ and give its units.

Exercise 3 (the multiplier as a common marginal value)

You have a fixed budget B to split between two activities with concave returns $R_1(x_1)$ and $R_2(x_2)$ (each $R'_i > 0$, $R''_i < 0$): $\max_{x_1, x_2} R_1(x_1) + R_2(x_2)$ subject to $x_1 + x_2 = B$.

- (a) Set up the Lagrangian and show the optimum equalizes marginal returns: $R'_1(x_1^*) = R'_2(x_2^*) = \lambda$.
- (b) What does λ say about the value of an extra unit of budget?
- (c) Suppose at some allocation $R'_1(x_1) > R'_2(x_2)$. Which way should you move a unit of budget, and why?